

Unit - I

Internet Architecture and Network Layer

(marks 12)

- Structure of Internet, Intranet, Role of Internet Service Provider (ISP) and Internet Corporation for Assigned Names and Numbers (ICANN)
- IPv4-Header format, IPv6 -Header format
- Subnet, subnet addressing and address masking, supernetting
- Address Mapping- Address Resolution Protocol (ARP) - Mapping logical to physical addresses, working and message format , Reverse Address Resolution Protocol (RARP) – Mapping physical to logical addresses working and message format

Structure of Internet

Today an Internet has a multi-backbone structure and these backbone networks run by different private corporations today.

- Several backbone networks, operated by private communication companies, provide global internet connectivity. These backbones are interconnected through peering points, enabling data exchange between them.
- Below the backbone level, provider networks utilize these backbone connections to offer internet services to end-users and organizations.
- At the lowest level, customer networks access the internet through the services offered by provider networks.

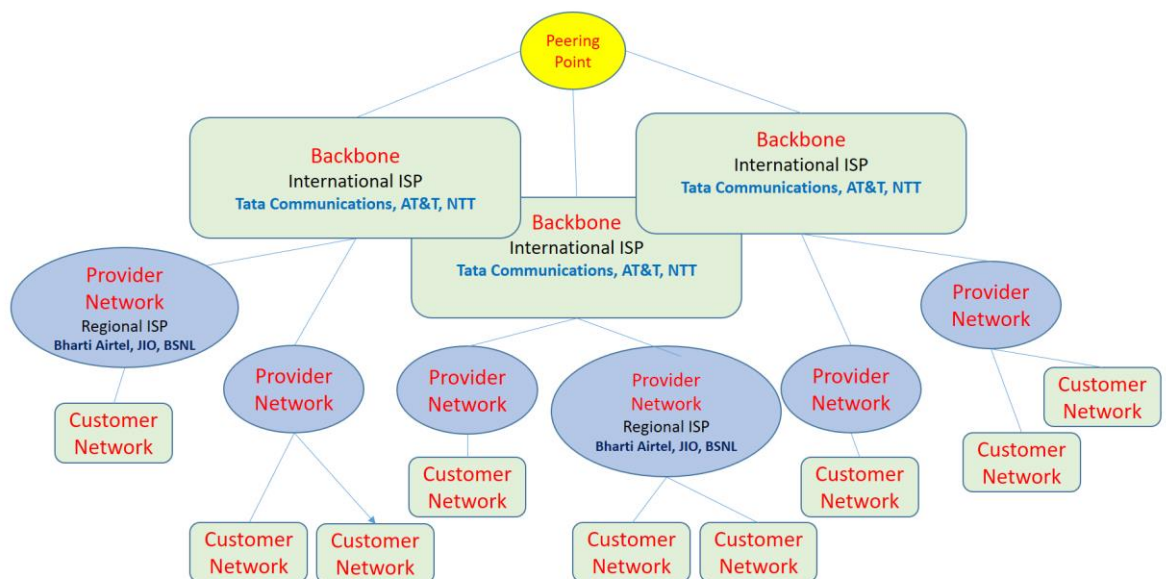


Fig1.1. Structure of Internet

Intranet

- A **private internal network within an organization** that uses TCP/IP protocol suite for communication.
- Uses internet technologies but restricted to internal access only.
- Provides services like internal email transfer, document sharing, internal employee chat system etc.

Role of ISP (Internet Service Provider)

- Provides internet access to users and businesses.
- Assigns public IP addresses.
- Offers services like email, domain hosting, and technical support.

- Any of these three entities (backbone, provider network, or customer network) can be called an Internet Service Provider or ISP. They provide services, but at different levels.

ICANN (Internet Corporation for Assigned Name and Numbers)

- Manages domain names, IP address allocation, and protocol identifiers.
- Ensures uniqueness and coordination of DNS across the internet.

Let's now understand the role of ICANN in the domain name registration process.

- Select a unique and relevant name for your website.
- Choose Top Level Domain based on your business type or region.
- Select a Domain Registrar
 - A domain registrar is a company accredited by ICANN to register domain names.
 - Popular registrars: GoDaddy, Namecheap, Google Domains, Bluehost.
- Go to the registrar's website.
 - Provide your contact details (name, email, phone, address).
 - These go into the WHOIS database (can be protected with privacy settings).
- Once payment is done, the registrar.
 - Registers the domain with ICANN
 - Updates the WHOIS record
 - Confirms registration via email

IPv4 (Internet Protocol version 4)

- Maximum size of IPv4 datagram : 65536B
- Maximum size of Header in IPv4: 60B
- Minimum size of Header in IPv4 : 20B

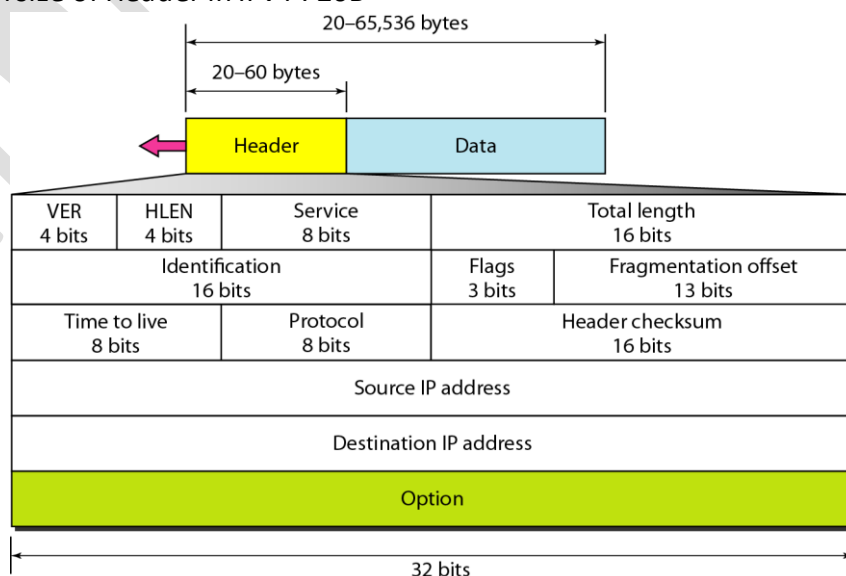


Fig. 1.2 IPv4 Datagram Header

IPV4 DATAGRAM HEADER FIELDS

VER (4 bits):

This 4-bit field defines the version of the IPv4 protocol. Currently the version is 4. However, version 6 (or IPng) may totally replace version 4 in the future.

HLEN (4 bits):

This 4-bit field defines the total length of the datagram header in 4-byte words. This field is needed because the length of the header is variable (between 20 and 60 bytes).

When there are no options, the header length is 20 bytes, and the value of this field is 5

($5 \times 4 = 20$). When the option field is at its maximum size, the value of this field is 15 ($15 \times 4 = 60$).

Total Length (16 bits):

Length in bytes of the datagram, including headers. Maximum datagram size is 65536 bytes.

Service Type (8 bits):

Service Type tells the network how to handle a packet, such as its priority, speed, and reliability.

Time to Live (8 bits):

Prevents a packet from traveling forever in a loop. Senders sets a value, which is decremented at each hop. If it reaches zero, packet is discarded.

Protocol:

Defines the upper layer protocol that uses the service of the IP layer

Header Checksum (16 bits):

Datagram is discarded in case of mismatch in checksum values

Source IP address (32 bits): Internet address of the sender.

Destination IP address (32 bits): Internet address of the destination.

Identification, Flags, Fragment Offset:

Used for handling fragmentation. (Discussed in detail later)

Options (variable width):

Can be used to provide more functionality to the IP datagram

IPV6 Packet Format

- The Ipv6 packet format is shown in figure-1.3. Each packet is composed of a mandatory base header followed by the payload.
- The payload consists of two parts: optional extension headers and data from an upper layer.
- The base header occupies 40 bytes, whereas the extension headers and data from upper layer contain up to 65,535 bytes of information.

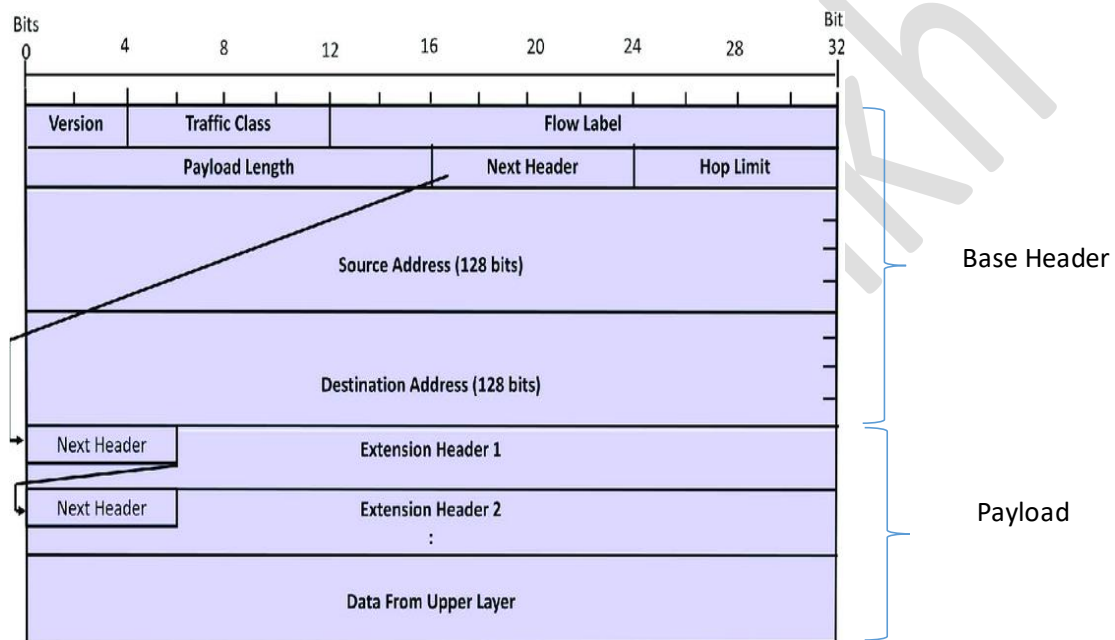


Fig. 1.3 IPv6 Packet Format

IPV6 Header Fields Description

Version (4-bits):

It represents the version of Internet Protocol, i.e. 0110.

Traffic Class (8-bits):

These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

Flow Label (20-bits):

This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

Payload Length (16-bits):

This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes and this field is set to 0.

Next Header (8-bits):

This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

Hop Limit (8-bits):

This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches 0 the packet is discarded.

Source Address (128-bits):

This field indicates the address of originator of the packet.

Destination Address (128-bits):

This field provides the address of intended recipient of the packet.

Subnetting

Subnetting is the process of dividing a large network into smaller, more manageable subnetworks (subnets).

Why Subnetting ?

- To improve network performance by reducing congestion.
- To enhance security by isolating network segments.
- To efficiently use IP addresses and reduce wastage.
- To simplify network management and troubleshooting.

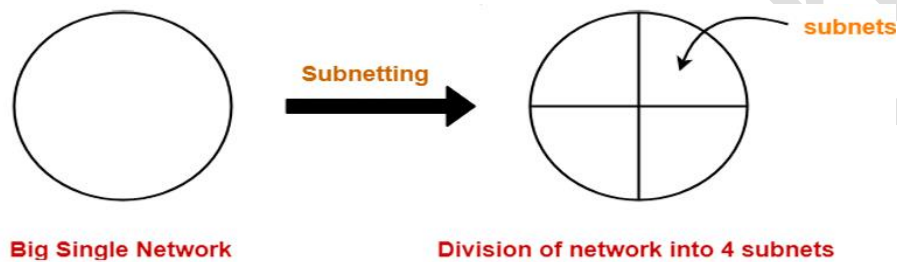


Fig. 1.4: Dividing a large network into small subnets (Subnetting)

Subnetting Types

Fixed Length Subnet Masking (FLSM)

Fixed length subnetting, which is also known as classful subnetting, divides the network into subnets in such a way that-

- All subnets are of the same size.
- All subnets have the same number of hosts.
- All subnets have the same subnet mask.

Variable Length Subnet Masking (VLSM)

Variable length subnetting, which is also known as classless subnetting, can be seen as follows in networking:

- All subnets are not of the same size.
- All subnets do not have the same number of hosts.

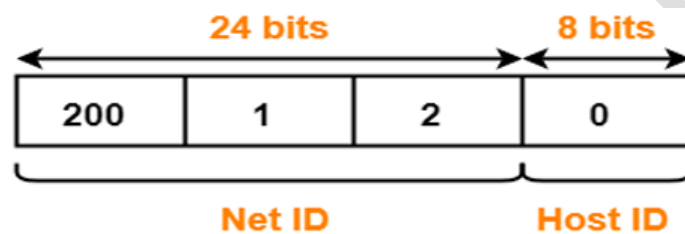
All subnets do not have the same subnet mask.

Problem1

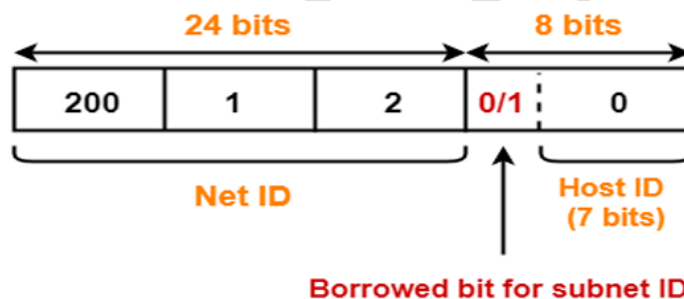
A company has been assigned the Class C network 200.1.2.0. It requires 2 equal-sized subnets. determine:

- The subnet masks.
- The network address of each subnet.
- The valid host ranges.
- The broadcast addresses.

SOLUTION



To represent 2 subnets we require 1bit ($2^1 = 2$). Borrow it from existing Host bits.



200.1.2.0

200.1.2.00000000 Subnet Mask 200.1.2.0 IP Range 200.1.2.0 – 200.1.2.127 Total Host : 128 Usable Host: 128 – 2 = 126 Network IP: 200.1.2.0 First IP: 200.1.2.1 Last IP: 200.1.2.126 Broadcast IP: 200.1.2.127	200.1.2.10000000 Subnet Mask 200.1.2.128 IP Range 200.1.2.128 – 200.1.2.255 Total Host : 128 Usable Host: 128 – 2 = 126 Network IP: 200.1.2.128 First IP: 200.1.2.129 Last IP: 200.1.2.254 Broadcast IP: 200.1.2.255
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Problem 2 (based on VLSM – Variable Sized Subnet Masks)

Design a Network with 3 Subnets (100, 30, and 10 Hosts), Design a network to support :

Subnet A: 100 hosts, Subnet B: 30 hosts, Subnet C: 10 hosts. Use 192.168.1.0/24 as the base network.

SOLUTION

100 hosts

Minimum 7 bit required for hosts addressing, therefore subnet mask will be 192.168.1.0/25 (total 128 IPs)

Network ID : 192.168.1.0 (Starts from the base)

First Valid IP : 192.168.1.1

:

Last Valid IP : 192.168.1.126

Broadcast Address: 192.168.1.127

30 hosts

Minimum 5 bit required for hosts addressing, therefore subnet mask will be 192.168.1.0/27 (total 32 IPs)

Network ID : 192.168.1.128 (Starts right after the previous subnet's broadcast address)

First Valid IP : 192.168.1.129

:

Last Valid IP : 192.168.1.158

Broadcast Address: 192.168.1.159

10 hosts

Minimum 4 bit required for hosts addressing, therefore subnet mask will be 192.168.1.0/28

Network ID : 192.168.1.160 (Starts right after the previous subnet's broadcast address)

First Valid IP : 192.168.1.161

:

Last Valid IP : 192.168.1.174

Broadcast Address: 192.168.1.175

Address Mapping

- Devices in a network are identified using two types of addresses: logical addresses (IP address) and physical addresses (MAC address in case of Ethernet Protocol). Physical addresses are used at the Data Link layer for node-to-node communication, while logical addresses are used at the Network layer for host-to-host communication.
- This means that delivering a packet to a host or router requires two levels of addressing: logical and physical. To achieve this, we must map a logical address to its corresponding physical address and vice versa, It can be done either by using static mapping or dynamic mapping methods.

Static Mapping

Static mapping involves creating a table that links each logical address to its corresponding physical address. This table is stored on every machine within the network. However, static mapping has certain limitations because physical addresses can change in the following situations:

- A device may replace its Network Interface Card (NIC), which results in a new physical address.
- In some LAN technologies, such as LocalTalk, the physical address may change every time the computer is restarted.

Dynamic Mapping

In dynamic mapping, when a computer knows another computer's logical address (IP address), it **uses a special protocol** to find its physical address (MAC address) or vice versa.

There are two main protocols used for this purpose:

- ARP (Address Resolution Protocol)
- RARP (Reverse Address Resolution Protocol)

Address Resolution Protocol (ARP)

ARP accepts a logical address from the IP protocol maps the address to the corresponding physical address and pass it to the data link layer.

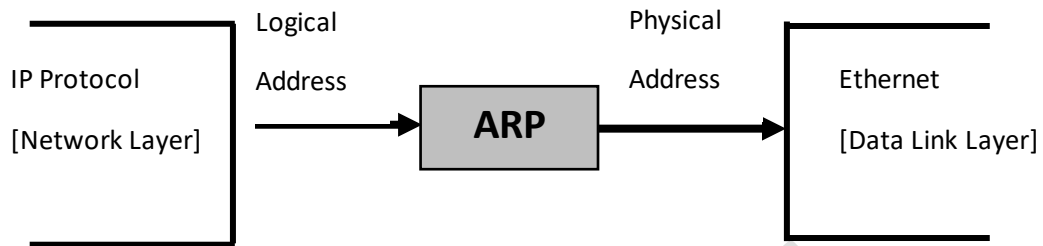


Fig. 1.5: ARP Position

Working

1. ARP Query Packet Broadcast

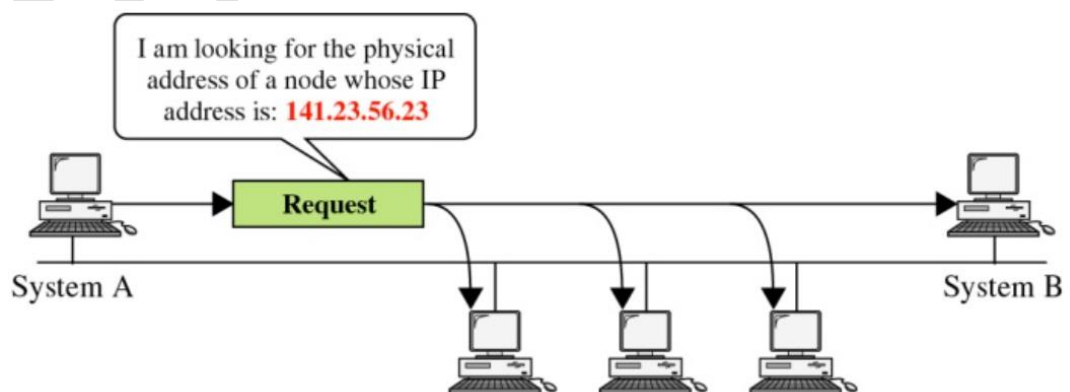
ARP Query Packet contains

- Senders IP Address
- Senders Physical Address
- Receivers IP Address (Whom Physical Address to be found)

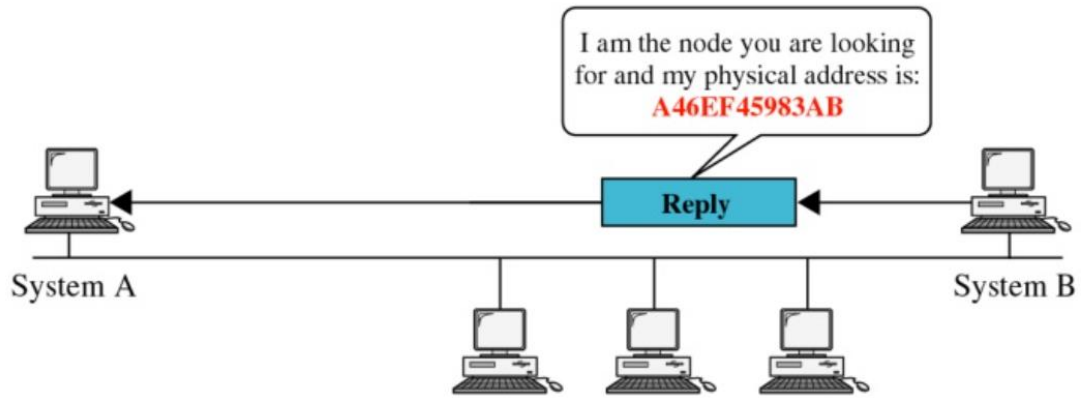
2. Every host or router on the network receives and examines the ARP request packet. However, only the device with the matching IP address recognizes the request and replies by sending an ARP response packet.

3. ARP Response Packet Unicast

The response packet includes the IP address and physical (MAC) address of the replying device. This packet is sent directly (unicast) to the original sender using the physical address provided in the ARP request.



ARP Request (Broadcast)



ARP Response (Unicast)

Fig. 1.6 ARP Working

ARP packet format

Hardware Type		Protocol Type
Hardware length	Protocol length	Operation Request 1, Reply 2
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled in a request)		
Target protocol address (For example, 4 bytes for IP)		

Fig. 1.7 ARP packet format

Hardware Type (16bit)

It define type of the network ARP is running, e.g. Ethernet and the integer value assigned is 1. Similarly, other hardware types as well assigned with integer number.

Protocol Type (16bit)

This field defines the protocol value, e.g. for IP protocol it is 0800₁₆.

Hardware Length (8bit)

This field defines length of physical address in terms of bytes, e.g. for Ethernet type network it is 48bit means 6 bytes.

Protocol Length (8bit)

This field defines length of logical address in terms of bytes, e.g. for IPv4 protocol it is 32bit means 4 bytes.

Operation Request (16bit)

This field defines the type of message is either a ARP Request (value is 1) or a ARP Response (value is 2).

Sender Hardware Address (variable length)

This field contains sender's physical address. For an Ethernet network, this field is 6 byte long.

Senders Protocol Address (variable length)

This field contains sender's logical address. For an IP protocol, this field is 32bit (4Byte) long.

Target Hardware Address (variable length)

This field contains target's physical address. For an Ethernet network, this field is 6 byte long. In case of ARP Request this field value is all 0's.

Target Protocol Address (variable length)

This field contains target's logical address. For an IPv4 protocol, this field is 4 byte long.

Reverse Address Resolution Protocol

- RARP is a network layer protocol used to map a physical address (MAC address) to a logical address (IP address). RARP is the reverse of ARP.
- Mainly used by diskless computers (which don't have storage to save IP addresses).
- It helps these computers automatically get their IP address from a RARP server at boot time.
- RARP is no longer used today for two main reasons:
 - RARP uses broadcasting, so a RARP server must be present on every local network, which is not efficient.
 - RARP only provides the IP address, but modern computers need more information to connect to a network—such as the subnet mask, default gateway, and DNS server.

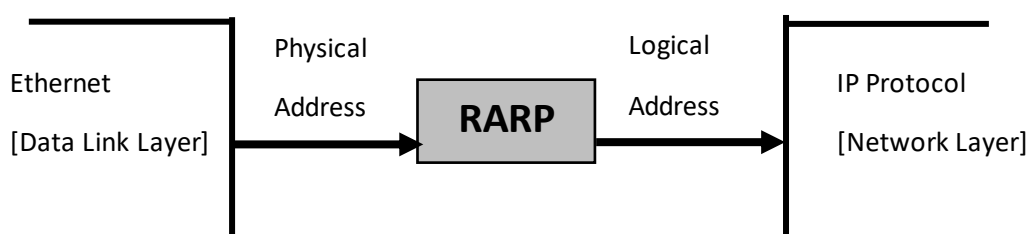


Fig. 1.8 RARP Position

RARP Working

1. A diskless machine knows only its MAC address, not its IP address.
2. When the machine boots up, it sends a RARP request as a broadcast on the network.
3. A RARP server (configured on the same LAN) receives the request and looks up the IP address that matches the MAC address.
4. The server then sends a RARP reply with the correct IP address to the requesting machine.

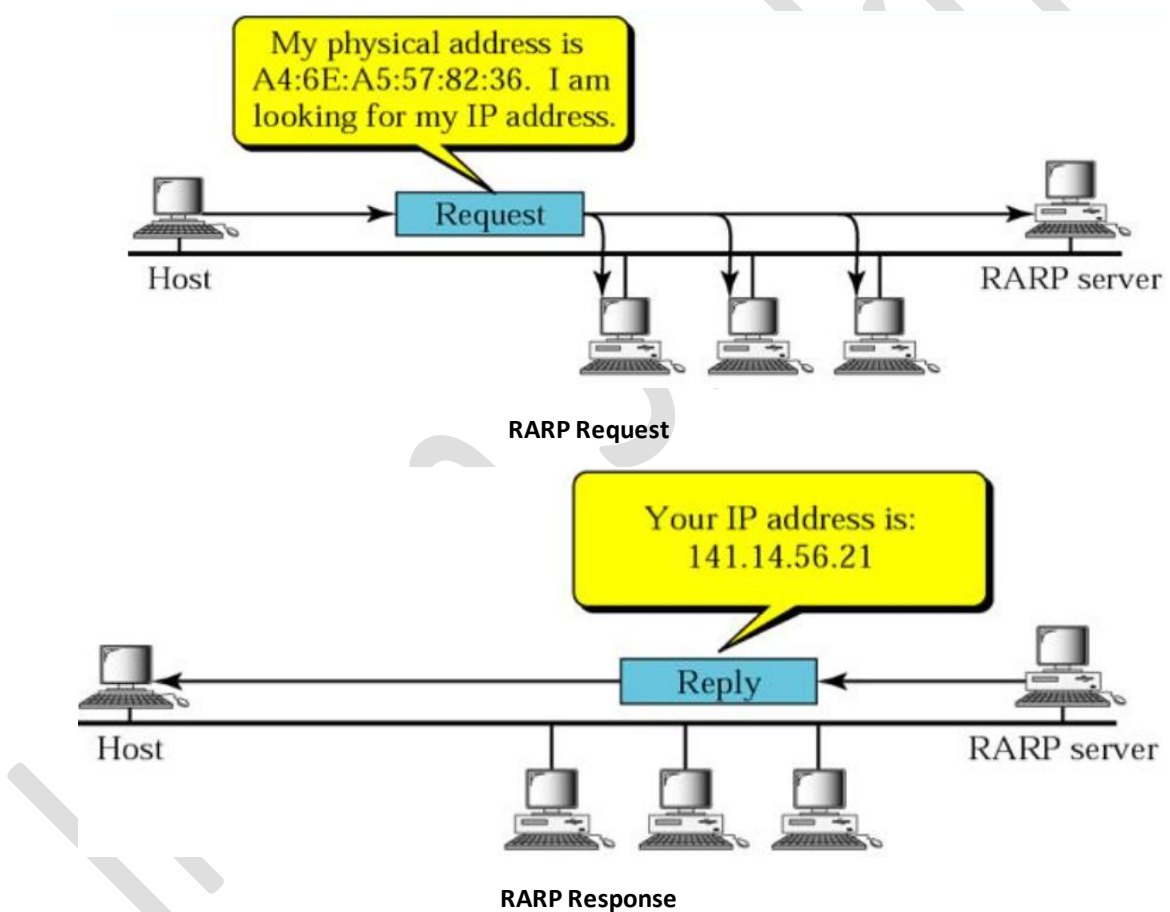


Fig. 1.9 RARP Working

RARP Packet Format

Hardware type		Protocol type
Hardware length	Protocol length	Operation Request 3, Reply 4
Sender hardware address (For example, 6 bytes for Ethernet)		
Sender protocol address (For example, 4 bytes for IP) (It is not filled for request)		
Target hardware address (For example, 6 bytes for Ethernet) (It is not filled for request)		
Target protocol address (For example, 4 bytes for IP) (It is not filled for request)		

Fig. 1.10 RARP packet format

Note:

The description of the fields in the RARP packet is the same as in ARP. The only difference is in the operation (opcode) field, where the value 3 indicates a RARP request and 4 indicates a RARP reply.